

Nibrexo OS AI

10 — Email Automation & Client Communication Specification v1.0

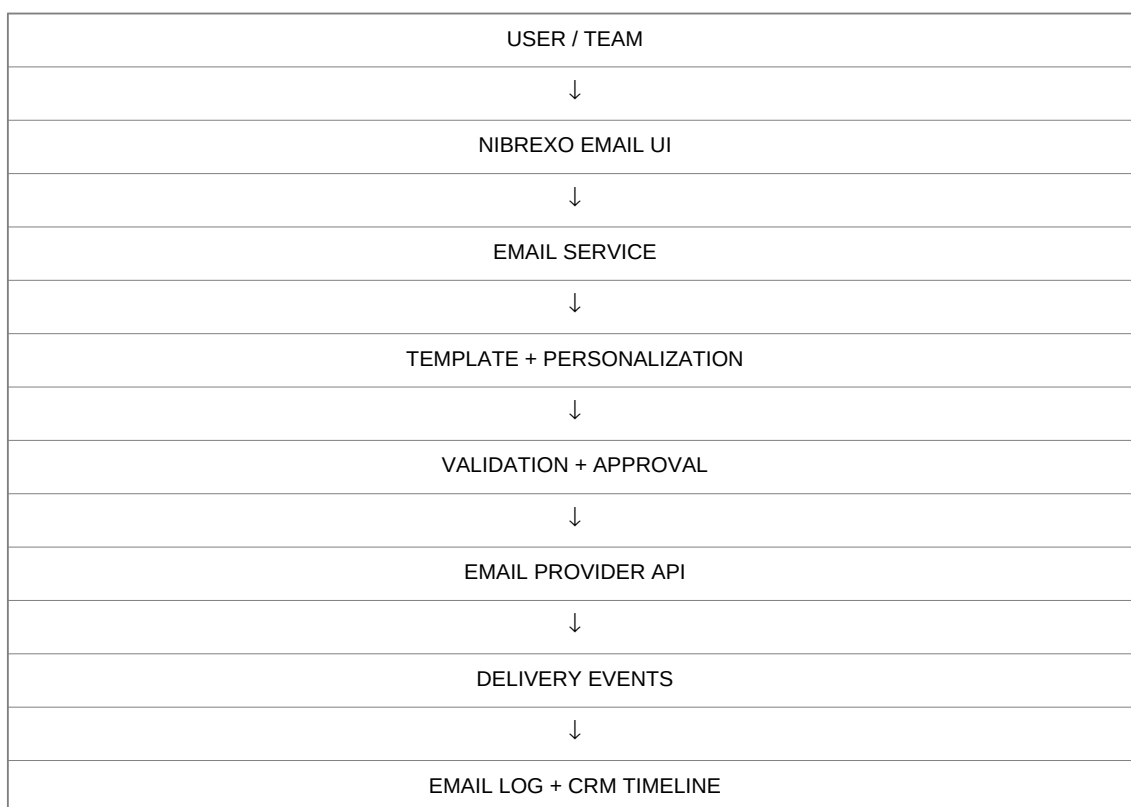
Production blueprint for client emails, reusable templates, AI-assisted personalization, automated sequences, approvals, delivery tracking and CRM communication history.

1. Objective

Nibrexo OS AI ka email module client communication ko centralized workspace mein manage karega. Templates, client data, AI personalization, approvals, external provider sending, follow-up sequences aur CRM history ek system mein connected honge.

Local/manual email sending product ka core nahi hoga. Production emails configured external email provider API ke through send honge.

2. Complete Email Architecture

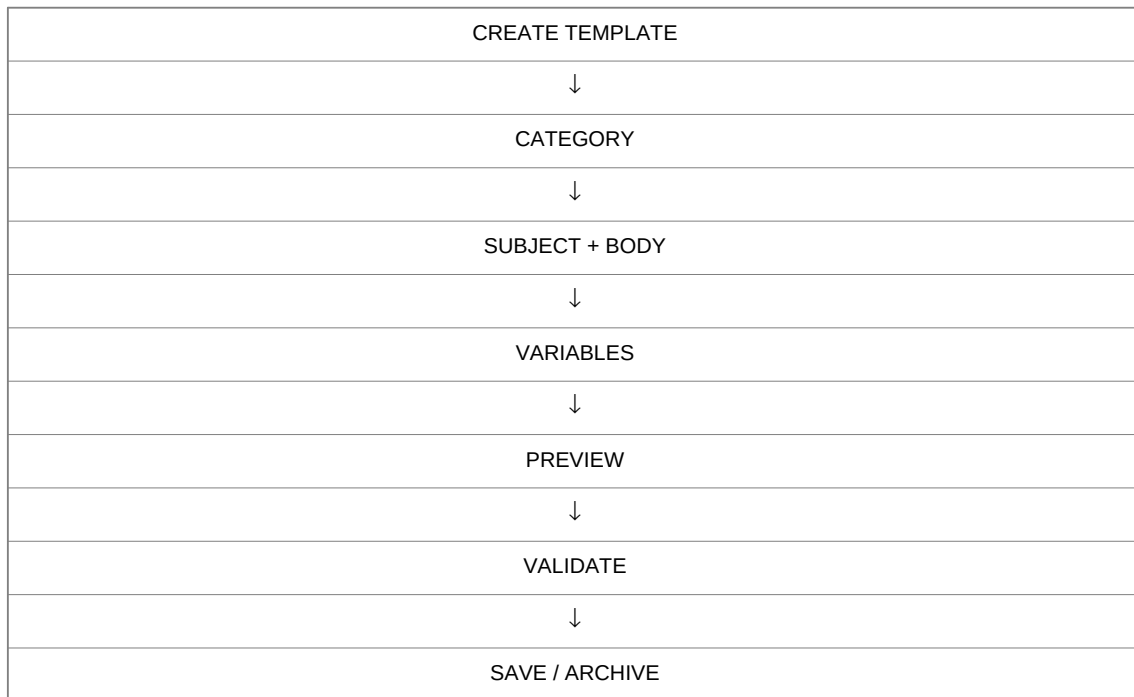


3. Core Modules

Module	Purpose
Templates	Reusable professional messages
Contacts	Client recipients
Composer	Create/edit emails
Personalization	Dynamic client/project data
Approval	Human review
Sequences	Automated follow-ups

Module	Purpose
Logs & Events	Send/delivery history
CRM Timeline	Communication history
Preferences	Communication controls

4. Template Flow

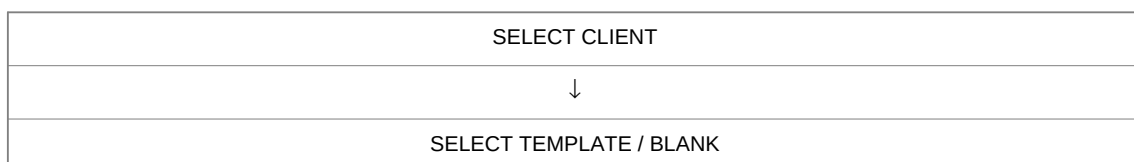


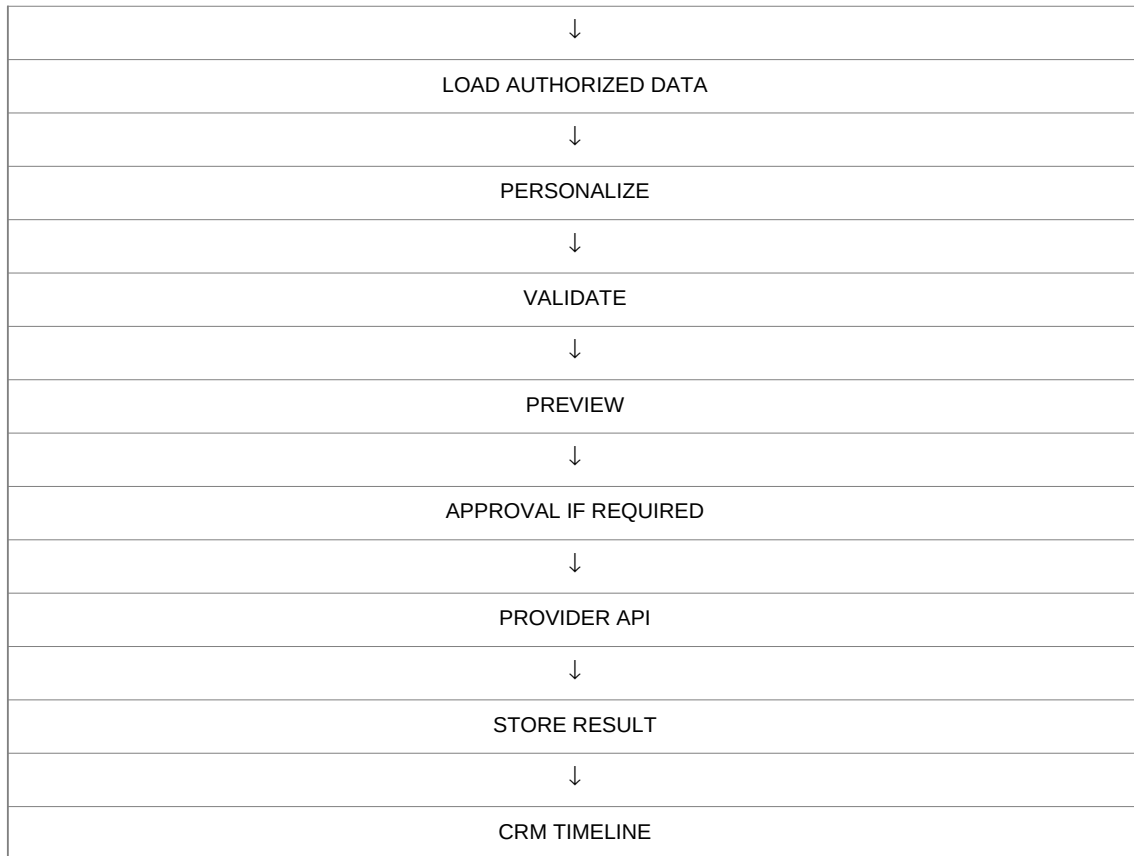
- Templates are organization-scoped.
- Variables are validated before sending.
- Archived templates cannot be accidentally selected.
- Template changes should be traceable.

5. Dynamic Variables

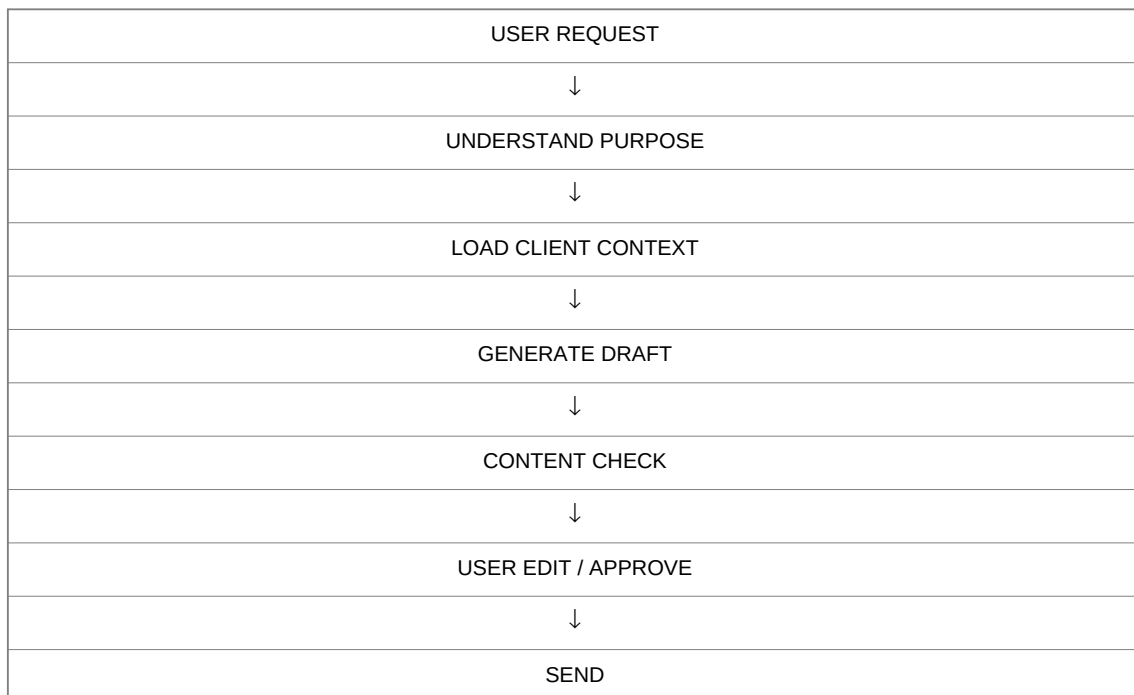
Variable	Example
Client	{{client.name}}
Company	{{client.company}}
Contact	{{contact.name}}
Project	{{project.name}}
Sender	{{user.name}}
Custom field	{{client.custom_field}}

6. Individual Email Flow



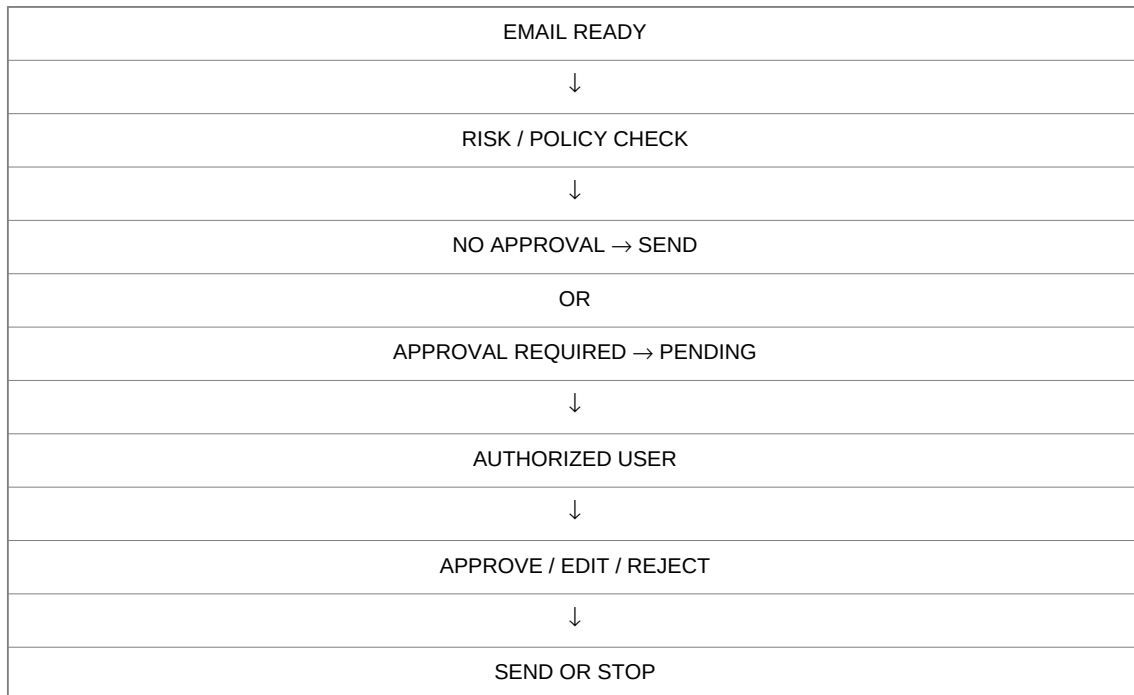


7. AI Email Assistant



- AI creates drafts and recommendations; it does not bypass authorization.
- AI output remains editable.
- Configured approval requirements always apply.
- Brand voice can be organization-configured.

8. Approval Engine



9. Automation Sequence Engine



10. Sequence Components

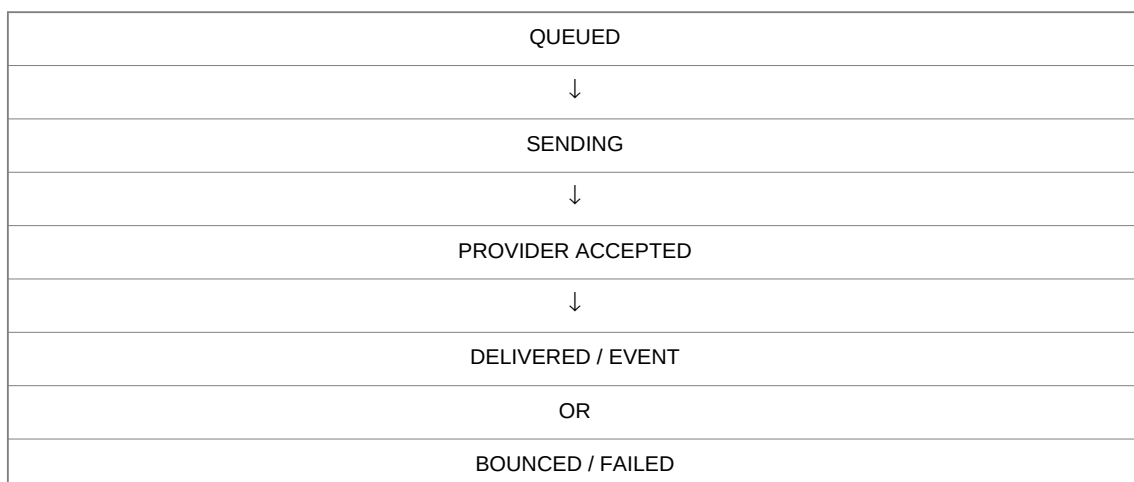
Component	Meaning
Trigger	Starts workflow
Step	One email/action
Delay	Wait period
Condition	Controls next path
Stop rule	Ends workflow
Status	Execution state
Retry policy	Safe temporary-failure handling

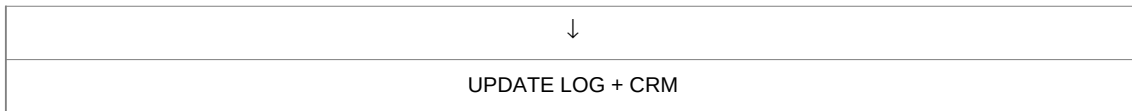
11. Example Follow-up



- Stop when a client replies where supported.
- Stop when user removes client.
- Respect applicable opt-out/preferences.
- Stop or pause when provider/account authorization is unavailable.
- Protect against duplicate execution.

12. Delivery Lifecycle

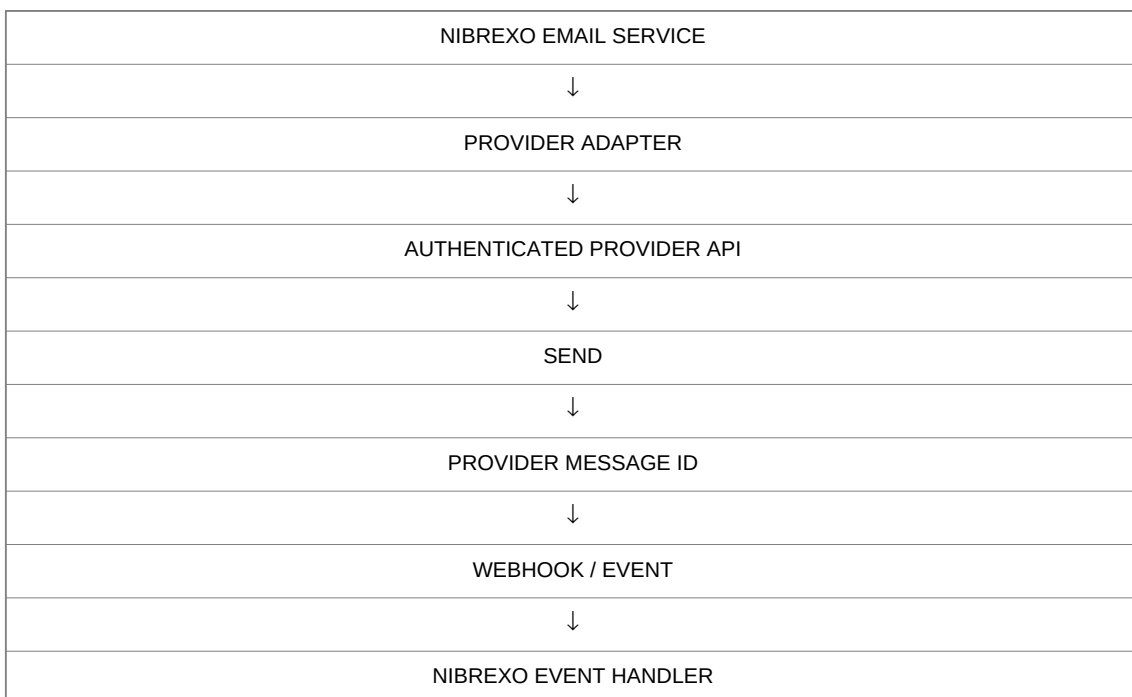




13. Email Status Model

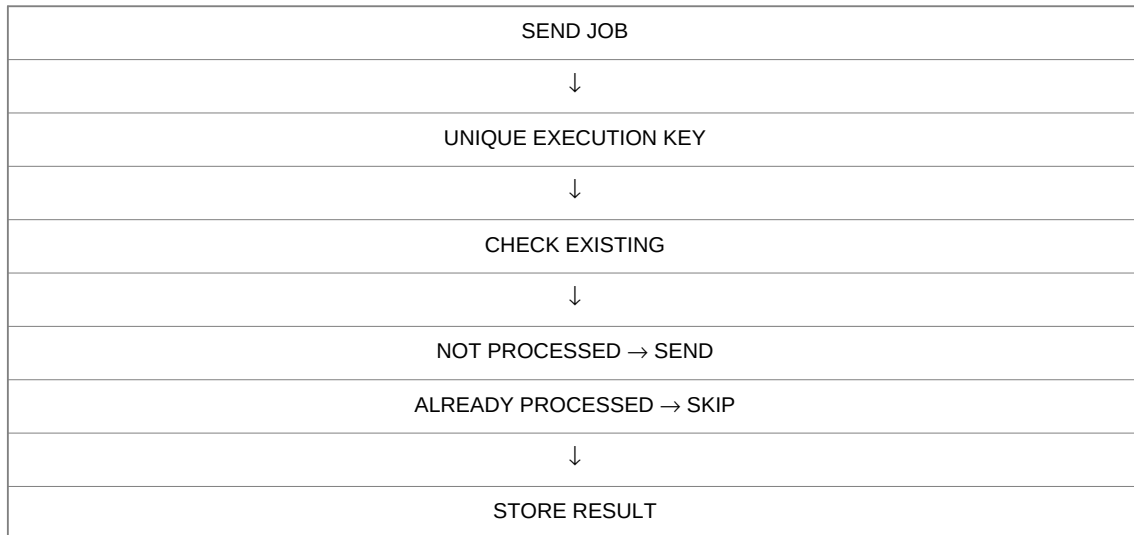
Status	Meaning
DRAFT	Not sent
PENDING_APPROVAL	Waiting
APPROVED	Ready
QUEUED	Waiting for worker
SENDING	Provider request
SENT	Provider accepted
DELIVERED	Confirmed where available
BOUNCED	Bounce event
FAILED	Send failure
CANCELLED	Cancelled

14. Provider Architecture



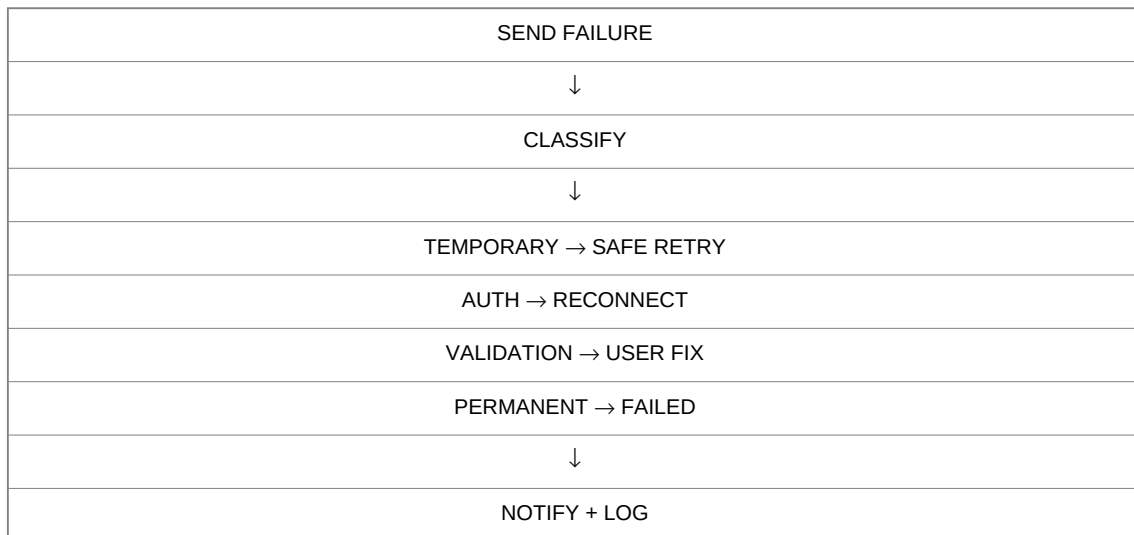
- Provider-specific logic stays inside an adapter.
- API credentials stay server-side.
- Provider message IDs are stored for tracing.
- Webhook authenticity is verified where supported.
- Provider capabilities must be verified before implementation.

15. Duplicate Protection



Retries must not accidentally create duplicate client emails.

16. Failure Handling



17. Database Entities

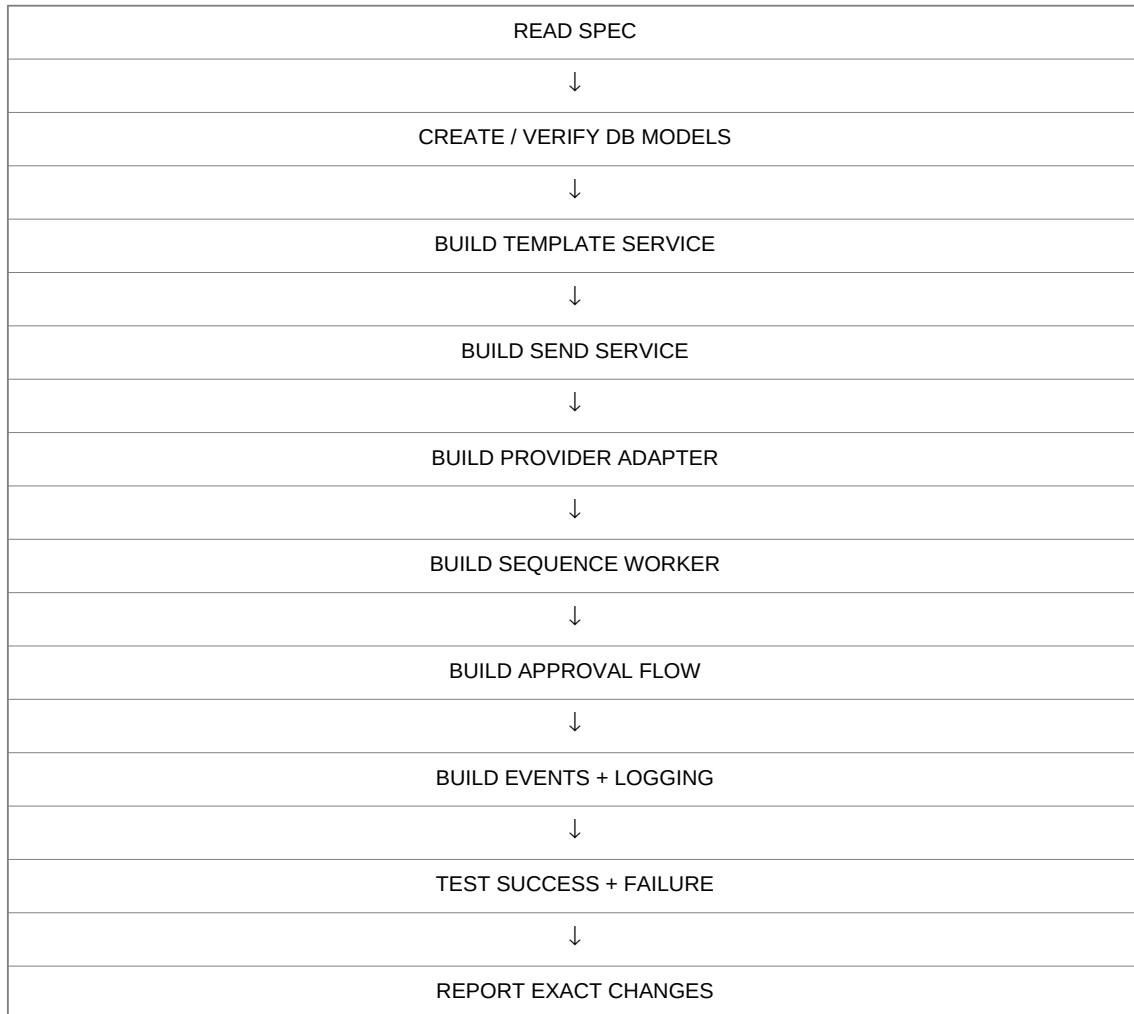
Table	Purpose
email_templates	Content, variables, versions
email_sequences	Automation definitions
email_steps	Sequence actions
email_jobs	Scheduled jobs
email_logs	Send records
email_events	Provider events
email_preferences	Communication preferences
client_activity	CRM timeline

18. Security Rules

- Provider/API secrets never enter frontend code.

- Client data is organization-scoped.
- Only authorized users can access communication.
- Sensitive logs avoid unnecessary secret data.
- Applicable communication preferences/opt-out requirements are respected.
- Important external sends are auditable.

19. LLM Arena Development Flow



20. Production Checklist

- Template CRUD complete.
- Variable validation complete.
- Composer/preview complete.
- Provider integration tested.
- Approval flow tested.
- Sequence scheduling tested.
- Duplicate protection tested.
- Delivery/bounce handling tested.
- CRM timeline tested.
- Permissions/RLS tested.

- Secrets protected.
- Failure/retry behavior tested.
- Audit logging enabled.

21. Product Rule

Nibrexo OS AI follows controlled email automation: the system may prepare drafts, templates, schedules and follow-ups, but it must not bypass permissions, configured approval, provider capabilities or communication preferences.

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